46-886 Machine Learning Fundamentals
Practice Exam

1. **Logistic Regression and Decision Trees**

In this problem, you will build models for an online retailer to predict the future demand of its repeat customers. The data, which is given in the file `demand.csv`, contains information on **1000** customers that made significant purchases in 2019. Imagining that it is December 31st, 2019, your goal will be to predict the variable `HighSpend`, which identifies customers who were "high" spenders in 2024. (Such binary classification can then be used to drive a targeted marketing campaign.) The other variables encode potentially relevant information about the customers, as described below.

Variable	**Description**
`HighSpend`	1 if the customer was a "high" spender in 2024, 0 if not
`Spend2019`	Dollars spent by the customer in 2019
`Age`	Customer's age in years (on December 31st, 2019)
`Cart`	Total dollar value of products added to the virtual shopping cart in 2019
`Device`	Primary device used. One of: `Mac`, `PC`, `Android`, `iOS`
`Email`	Email account provider. One of: `Gmail`, `Yahoo`, `Other`
`Income`	Customer's annual income
`Payment`	Primary payment method. One of: `Debit`, `Credit`, `Paypal`
`Premium`	Percentage of purchases in 2019 that were premium products
`Promotion`	Percentage of purchases in 2019 that were discounted
`Region`	One of: `Northwest`, `Southwest`, `Midwest`, `Northeast`, `Southeast`
`Sessions`	Number of online shopping sessions in 2019
`Time`	Number of minutes spent online shopping in 2019
`Clothing`	1 if one of the purchases in 2019 came from the Clothing category, 0 if not
`Books`	1 if one of the purchases in 2019 came from the Books category, 0 if not
`Electronics`	1 if one of the purchases in 2019 came from the Electronics category, 0 if not
`Home`	1 if one of the purchases in 2019 came from the Home category, 0 if not
`Garden`	1 if one of the purchases in 2019 came from the Garden category, 0 if not
`Pet`	1 if one of the purchases in 2019 came from the Pet category, 0 if not
`Grocery`	1 if one of the purchases in 2019 came from the Grocery category, 0 if not
`Health`	1 if one of the purchases in 2019 came from the Health category, 0 if not
`Outdoors`	1 if one of the purchases in 2019 came from the Outdoors category, 0 if not
`Automotive`	1 if one of the purchases in 2019 came from the Automotive category, 0 if not
`Beauty`	1 if one of the purchases in 2019 came from the Beauty category, 0 if not
`Sports`	1 if one of the purchases in 2019 came from the Sports category, 0 if not
`Tools`	1 if one of the purchases in 2019 came from the Tools category, 0 if not

Before proceeding to the questions, perform the following pre-processing steps:

- Apply one-hot encoding to the variables `Device`, `Email`, `Payment`, and `Region`.
- Divide the data into a training set consisting of the first 700 rows, and a test set containing the last 300 rows.

(a) Train an unregularized *Logistic Regression* model using all of the features. Report your model's out-of-sample AUC.

(b) Calculate

$$\frac{\text{Predicted Odds of Customer X being a High Spender in 2024}}{\text{Predicted Odds of Customer Y being a High Spender in 2024}}$$

where X and Y are identical except X had spent 3 fewer hours online shopping and resides in the Northeast in comparison to Y who resides in the Midwest.

(c) Now train a *Decision Tree* with maximum depth 2, with default hyperparameters otherwise. Plot this tree and include the figure in your report.

(d) Describe in detail the type of customer that your decision tree predicts is most likely to be a high spender.

(e) Train another *Decision Tree*, this time using 10-fold cross-validation to select the tuning parameter `min_samples_split` (please choose an appropriate range, please use scikit-learn's default value for `max_depth`). Report your model's out-of-sample AUC, the optimal value of `min_samples_split` thus obtained, and the range of `min_samples_split` the search was carried over.

(f) Compute the out-of-sample confusion matrix for your decision tree model from part (e), using a threshold of 0.1 to convert continuous values to binary predictions. Report the True Positive Rate and True Negative Rate.

2. **Skipping the Pink Slip**

Here's a scenario inspired by ≈ real events. You are a senior data scientist at Zeelow Inc. It's your first day at work, and you are given a dataset containing house prices (what we want to predict) and square footage of the house (just 1 feature).

The dataset is in `zeelow.csv`. There is no test/train split for this question.

(a) Your boss (who leads Product Strat) wants to "understand" how house prices vary across sizes, and ask you to train a linear regression model with houses prices as the dependent variable. Report the R2 score and the slope term.

(b) Hm.. you realize that something's off. Presenting the finding as is wouldn't be great for your job security.

You ask around for how the data was collected. You learn that it was aggregated by a survey agency that operates across various countries, which you know could mean that the houses surveyed could be from countries with drastically different per-capita GDP. For example, Country A (which you suspect was surveyed) is largely agrarian, and although the people on average are the least well off there (among all countries in the world), they live in large houses. Country B (which you suspect was also surveyed) is a landlocked country with limited land availablity (with the smallest house sizes in the world), but it is a global financial hub – real estate prices there are quite a bit higher than anywhere else. But in any case, there's no way to reach out to the survey agency.

Can you figure out which houses are likely to belong to same country? How many countries are represented in the dataset? (When in doubt, choose a smaller number.) Calculate the average price of houses in County B (which is the most well off), average size of houses in Country A (the agrarian one), and the number of houses that you suspect are neither in Country A nor in Country B.

(c) Assuming you figured out the last part, you breathe a sigh of relief. Now, you know which houses belong to which country (or technically, at least which pairs belong in the same). You split your dataset so that all samples in a partitioned dataset belong to the same country, and run linear regression on each separately.

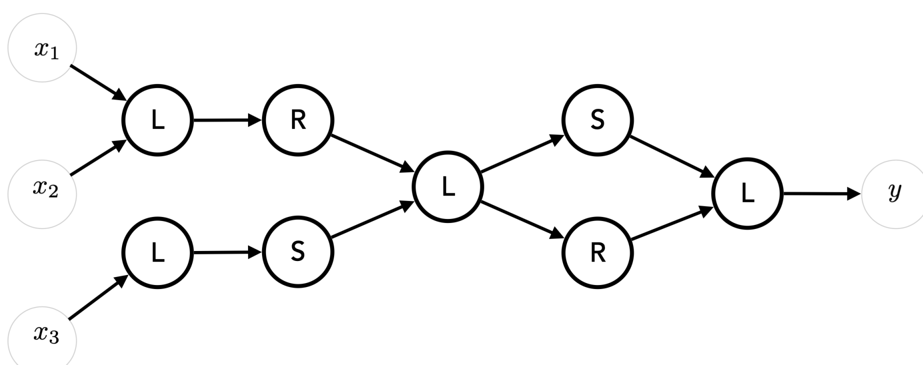
Report the predicted increase in house prices with each additional unit of area for Country A (agrarian). Same for Country B (finance hub). Also, report the R2 for the regression instance for Country A (agrarian); no need for B.

(d) Explain in words the source of the discrepancy, or lack of it, between your findings concerning the relationship between price and size in the parts (a) and (c) above. Set aside your ML knowledge, how would you explain it to a lay person?

3. **Neural Networks**

First, some general questions about neural networks:

(a) The following is a small neural network. Three variables x_1, x_2, x_3 are input, and the letters L, R, and S represent linear, ReLU, and sigmoid nodes, respectively. The output, y, is thus a function of x_1, x_2, x_3. How many trainable parameters does this neural net have?



Believe it or not, there is yet another sibling to the famous MNIST dataset. This one is called **EMNIST** (the "E" stands for Extended), and contains handwritten letters of the alphabet. Tthe images in EMNIST are 28-by-28 pixels in grayscale, there are 60,000 training samples, and there are 10,000 test samples. Here are a few example images from the dataset:

You can load the data by running the following command:

```
X_train,X_test,y_train,y_test = pickle.load(open(‘emnist.p’,‘rb’))
```

If you inspect `y_train` or `y_test`, you will see that the labels take values between 0 and 25. Logically, 0 corresponds to the letter A, 1 corresponds to the letter B, and so on.

(b) Consider a neural network with the following architecture:

- a linear layer with 256 nodes, followed by
- a ReLU layer, followed by
- a linear layer with 128 nodes, followed by
- a ReLU layer, followed by
- a linear layer with 26 nodes

Train this network until you are sufficiently confident it has converged, and report the out-of-sample accuracy. In your writeup, include screenshots of both the model itself (in code), and your accuracy.

(c) How many (trainable) parameters does the network from part (c) have? Why did we choose (the output dimension of) the last layer to be 26 in width?

(d) Consider a neural network with the following architecture:

- a convolutional layer with 32 nodes, each applying a 5-by-5 filter, followed by
- a ReLU layer, followed by
- a MaxPool layer with 2-by-2 window, followed by
- a convolutional layer with 16 nodes, each applying a 5-by-5 filter, followed by
- a ReLU layer, followed by
- a MaxPool layer with 2-by-2 window, followed by
- a linear layer with 128 nodes, followed by
- a ReLU layer, followed by
- a linear layer with 26 nodes

Train this network until you are sufficiently confident it has converged, and report the out-of-sample accuracy. In your writeup, include screenshots of both the model itself (in code), and your accuracy.

(e) How many (trainable) parameters does each layer in the network from part (e) have?